

Quiz 4

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (8 points) Let $\mathbf{x}(t) = e^{-t}\mathbf{i} - \sqrt{2}t\mathbf{j} + e^t\mathbf{k}$, $t \in \mathbb{R}$, that defines a path in $\mathbb{R}^3$.

(a) Find the parametric equations for the line tangent to the path at the point with $t = 0$.

$$\begin{aligned} \mathbf{x}(t) &= (e^{-t}, -\sqrt{2}t, e^t) \Rightarrow \mathbf{x}(0) = (1, 0, 1) \\ \mathbf{x}'(t) &= (-e^{-t}, -\sqrt{2}, e^t) \Rightarrow \mathbf{x}'(0) = (-1, -\sqrt{2}, 1) \\ \Rightarrow \text{the parametric equation for the tangent line} \\ &\text{are given by} \end{aligned}$$

$$\begin{aligned} \ell(t) &= \mathbf{x}(0) + t\mathbf{x}'(0) = (1, 0, 1) + t(-1, -\sqrt{2}, 1) \\ &= (1-t, -\sqrt{2}t, 1+t) \quad \left(\text{or } \begin{cases} x(t) = 1-t \\ y(t) = -\sqrt{2}t \\ z(t) = 1+t \end{cases} \right) \end{aligned}$$

(b) Find the length of the arc defined by the path with $0 \leq t \leq 1$.

$$\begin{aligned} \|\mathbf{x}'(t)\| &= \sqrt{(-e^{-t})^2 + (-\sqrt{2})^2 + (e^t)^2} \\ &= \sqrt{e^{-2t} + 2 + e^{2t}} = \sqrt{(e^{-t} + e^t)^2} = e^{-t} + e^t \end{aligned}$$

$$\begin{aligned} \Rightarrow L &= \int_0^1 \|\mathbf{x}'(t)\| dt \\ &= \int_0^1 (e^{-t} + e^t) dt \\ &= (-e^{-t} + e^t) \Big|_{t=0}^{t=1} \\ &= \left(-\frac{1}{e} + e\right) - (-1 + 1) = e - \frac{1}{e} \end{aligned}$$

Problem 2. (4 points) Compute the divergence of $\mathbf{F}$, $\nabla \cdot \mathbf{F}$, where $\mathbf{F} = x^2\mathbf{i} - xe^{2y}\mathbf{j} + 3xyz\mathbf{k}$.

$$\begin{aligned} \nabla \cdot \mathbf{F} &= \nabla \cdot (x^2, -xe^{2y}, 3xyz) \\ &= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(-xe^{2y}) + \frac{\partial}{\partial z}(3xyz) \\ &= 2x - 2xe^{2y} + 3xy \\ &= x(2 - 2e^{2y} + 3y) \end{aligned}$$

Problem 3. (8 points) Let $\mathbf{x}(t) = (\frac{3}{2} \sin 2t, 4t, \frac{3}{2} \cos 2t)$, $t \in \mathbb{R}$, that defines a path in $\mathbb{R}^3$. Find $\mathbf{T}(t)$, $\mathbf{N}(t)$, $\mathbf{B}(t)$ and the curvature $\kappa(t)$ of the path.

$$\mathbf{x}'(t) = (3 \cos 2t, 4, -3 \sin 2t)$$

$$\|\mathbf{x}'(t)\| = \sqrt{(3 \cos 2t)^2 + 4^2 + (-3 \sin 2t)^2} = \sqrt{9 \cos^2 2t + 16 + 9 \sin^2 2t} = \sqrt{25} = 5$$

$$\Rightarrow \mathbf{T}(t) = \frac{\mathbf{x}'(t)}{\|\mathbf{x}'(t)\|} = \frac{1}{5} (3 \cos 2t, 4, -3 \sin 2t) = (\frac{3}{5} \cos 2t, \frac{4}{5}, -\frac{3}{5} \sin 2t)$$

$$\mathbf{T}'(t) = (-\frac{6}{5} \sin 2t, 0, -\frac{6}{5} \cos 2t)$$

$$\|\mathbf{T}'(t)\| = \sqrt{(-\frac{6}{5} \sin 2t)^2 + 0^2 + (-\frac{6}{5} \cos 2t)^2} = \frac{6}{5}$$

$$\Rightarrow \mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \frac{1}{\frac{6}{5}} (-\frac{6}{5} \sin 2t, 0, -\frac{6}{5} \cos 2t) = (-\sin 2t, 0, -\cos 2t)$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{3}{5} \cos 2t & \frac{4}{5} & -\frac{3}{5} \sin 2t \\ -\sin 2t & 0 & -\cos 2t \end{vmatrix}$$

$$= \hat{i} \left(-\frac{4}{5} \cos 2t \right) - \hat{j} \left(-\frac{3}{5} \cos^2 2t - \frac{3}{5} \sin^2 2t \right) + \hat{k} \left(\frac{4}{5} \sin 2t \right)$$

$$= \left(-\frac{4}{5} \cos 2t, \frac{3}{5}, \frac{4}{5} \sin 2t \right)$$

$$\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{x}'(t)\|} = \frac{\frac{6}{5}}{5} = \frac{6}{25}$$